

TRONTEK

RECHARGEABLE SEALED MAINTENANCE FREE LEAD ACID BATTERIES



12V 5Ah

6v/4.5ah to 12v/100ah.
(2v & 4v Models Available on request)

Preferred by leading OEMs in
India and All over World

Trontek SMF Batteries

(Manufactured in the most modern & contemporary Taiwan-Made Plant)

Have all the winning features:

- Lead-Calcium grid for extended life
- High discharge rate capacity
- Deep discharge recovery
- Low self discharge & long service life
- No memory effect & high performance battery
- Long life & hassle free power

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TRONTEK GROUP

Q1. How is the voltage stepped down in mobile chargers without using a transformer?

R.C. Mittal

Kota

A1. Chargers for mobile phones use transformerless power supply. Mains voltage is brought down to a lower AC voltage by using a capacitive impedance in series, usually parallel combination of a high-wattage resistor and a capacitor. In other words, mains voltage is stepped down from 220V to 5-6V by this arrangement instead of a step-down transformer. The voltage is rectified using conventional half-wave or full-wave configuration. Regulation is achieved by using a zener diode followed by smoothen-

nels and 96kHz, 32-bit quality for up to eight channels.

Intel high-definition audio is designed specifically for a high-quality, multichannel audio experience. Newer audio and video encoding/decoding algorithms also enable a higher-quality listening experience.

HD Audio features multistreaming capabilities that allow users to send two or more different audio streams to different locations at the same time, from the same PC.

Q3. What are ultracapacitors and how do they work?

Chaitanya Deshpande

Through e-mail

A3. Ultracapacitors, also called electric double-layer capacitors,

are electrochemical capacitors with an unusually high energy density compared to common capacitors. They hold hundreds of times more energy per unit

volume or mass than conventional capacitors.

Typically, ultracapacitors have values thousands of times greater than high-capacity electrolytic capacitors. For instance, a typical D-cell-sized electrolytic capacitor will have a capacitance in the range of tens of milli-Farads. The same-size electric double-layer capacitor would have a capacitance of several Farads—an improvement of about two or three orders of magnitude in capacitance, but usually at a lower working voltage. Larger, commercial electric double-layer capacitors have capacities as high as 5000 Farads.

Ultracapacitors store electricity by physically separating positive and negative charges, unlike batteries that do so chemically. Energy is stored electrostatically by polarising an electrolytic solution. Though the ultracapacitor is

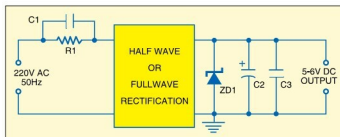


Fig. 1: Suggested circuit for a transformerless DC power supply

ing through an electrolytic capacitor of a large value.

Fig. 1 shows the suggested circuit for a transformerless DC power supply. It's a high AC voltage circuit, so one needs to observe safety precautions.

Q2. What is HD Audio?

Shyam Sunder Prajapat

Through e-mail

A2. HD Audio (or Azalia) is a specification released by Intel for delivering high-definition audio that is capable of playing back more channels at higher quality than previous integrated audio codecs like AC97. In addition, Intel high-definition audio supports the latest audio content. By enabling enhanced usage models, the hardware based on Intel HD Audio specification is capable of delivering 192kHz, 32-bit quality for two chan-